

Aligned Macaev Operators Lift Through the Canonical ELP Twist

Automated mathematics run `fa_banach_001`, lane 9

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Status. Candidate partial result, likely valid. The universal Macaev-ideal and $O(1/\log n)$ questions remain open.

1 Source question and scope

Cabello Sánchez and García ask on page 5 of [1]:

Is every operator in the Macaev ideal liftable?

Here the Macaev ideal consists of compact operators T with $\sum_{j \geq 1} s_j(T)/j < \infty$.

Note that $\|F\|$ may depend not only on the norms of the summands, but also on the Banach-Mazur distance between $\mathcal{H} \oplus_\phi \mathcal{H}$ and its Banach envelope which, in turn, depends only on $Q(\phi)$. $\square$

It is clear that an operator belongs to *some* $\mathcal{S}_p$ if and only if its singular numbers are $O(n^{-\alpha})$ for *some* $\alpha > 1$, which does not necessarily agree with $1/p$. An ideal that contains all the Schatten classes and is commonly regarded as reasonably small is the Macaev ideal $\mathcal{S}_\omega$ of those operators whose singular values satisfy $\sum_n s_n/n < \infty$, with the obvious norm, see [15, 8]. This ideal appears naturally in a surprisingly wide variety of situations, problems, results and applications, as a quick internet search reveals.

QUESTION. Is every operator in the Macaev ideal liftable?

Figure 1: The question in arXiv:2112.03852, page 5.

The theorem below proves the question for operators that are scalar on the canonical finite-dimensional blocks, and for lifting through the canonical Enflo–Lindenstrauss–Pisier (ELP) twisted Hilbert sequence. It does not prove universal liftability.

2 Definitions

For a homogeneous quasilinear map $\Phi : E \rightarrow F$ between finite-dimensional Hilbert spaces, put

$$\delta(\Phi) = \inf_{L \in \text{Lin}(E, F)} \sup_{x \neq 0} \frac{\|\Phi(x) - Lx\|}{\|x\|}.$$

The canonical ELP construction uses $H_n = \ell_2^{3^n}$ and maps $\phi_n : H_n \rightarrow H_n$. Write elements of H_{n+1} as triples $(x, y, z) \in H_n^3$, let $P_n(x, y, z) = (x, y, 0)$, and set

$$\phi_1(x, y, z) = \left(0, 0, \frac{x|y|}{(|x|^2 + |y|^2)^{1/2}} \right),$$

with value 0 when the denominator vanishes, and

$$\phi_{n+1}(x, y, z) = \left(\phi_n(x), \phi_n(y), \frac{P_n x \|P_n y\|}{(\|P_n x\|^2 + \|P_n y\|^2)^{1/2}} \right).$$

This is the construction restated in [2, Section 2.3]. On the algebraic direct sum in $H = (\bigoplus_{n \geq 1} H_n)_2$, define

$$\Omega_{\text{ELP}}(x_n)_n = (\phi_n(x_n))_n.$$

Its completion gives the canonical ELP twisted Hilbert sequence

$$0 \longrightarrow H \longrightarrow H \oplus_{\Omega_{\text{ELP}}} H \longrightarrow H \longrightarrow 0.$$

For a bounded scalar sequence $a = (a_n)$, let $D_a = \bigoplus_{n \geq 1} a_n I_{H_n}$.

3 Idea of the proof

For a block direct-sum twist, lifting a block scalar operator is exactly a uniform finite-block linearization problem: the obstruction on block n is $|a_n| \delta(\phi_n)$. The explicit ELP recursion has size at most $\sqrt{n}/2$, while the classical complementability lower bound forces the same $\sqrt{n}$ order. Thus the exact lifting condition is $\sup_n \sqrt{n} |a_n| < \infty$.

The endpoint consequence is then a counting observation. The n -th block has dimension 3^n , so it contributes 3^n copies of $|a_n|$ to the singular sequence. Macaev summability forces $n|a_n|$ to be bounded, which is stronger than the required $\sqrt{n}|a_n|$ bound.

4 The partial result

Theorem 1. *Let $\delta_n = \delta(\phi_n)$. For every bounded scalar sequence a , the following are equivalent:*

1. D_a lifts through the canonical ELP twisted Hilbert sequence;
2. $\sup_n |a_n| \delta_n < \infty$;
3. $\sup_n \sqrt{n} |a_n| < \infty$.

Consequently, D_a lifts through the ELP sequence whenever either

$$\sum_{j \geq 1} \frac{s_j(D_a)}{j} < \infty \quad \text{or} \quad s_j(D_a) = O\left(\frac{1}{\log(j+1)}\right).$$

Proof. We first prove the exact block criterion. Suppose D_a has a bounded lifting. In the quasilinear model its restriction to the algebraic direct sum has the form

$$x \longmapsto (Lx, D_a x)$$

for a linear map L . Boundedness says that $\|Lx - \Omega_{\text{ELP}}(D_a x)\| \leq C\|x\|$. Restricting to H_n and projecting the first coordinate onto H_n gives a linear map $L_n : H_n \rightarrow H_n$ such that

$$\|L_n x - a_n \phi_n(x)\| \leq C\|x\|.$$

Homogeneity of ϕ_n therefore gives $|a_n| \delta_n \leq C$.

Conversely, suppose $M = \sup_n |a_n| \delta_n < \infty$. Choose linear $L_n : H_n \rightarrow H_n$ so that

$$|a_n| \|\phi_n - L_n\| \leq M + 1.$$

(When $a_n = 0$, take $L_n = 0$.) On the algebraic direct sum set $L = \bigoplus_n a_n L_n$. Then

$$\|Lx - \Omega_{\text{ELP}}(D_a x)\|^2 = \sum_n |a_n|^2 \|L_n x_n - \phi_n(x_n)\|^2 \leq (M + 1)^2 \|x\|^2.$$

Thus $x \mapsto (Lx, D_a x)$ is bounded into the twisted-sum completion and extends to a lifting of D_a . This proves $(1) \Leftrightarrow (2)$.

We next estimate δ_n . The elementary inequality

$$\frac{r^2 s^2}{r^2 + s^2} \leq \frac{r^2 + s^2}{4}$$

and the recursion give

$$\|\phi_1\| \leq \frac{1}{2}, \quad \|\phi_{n+1}\|^2 \leq \|\phi_n\|^2 + \frac{1}{4}.$$

Hence $\delta_n \leq \|\phi_n\| \leq \sqrt{n}/2$.

For the reverse estimate, the ELP finite block satisfies the classical lower bound that its canonical copy of H_n is at least $c\sqrt{n}$ -complemented; see [2, Section 2.3], which cites the original construction [3]. If $L : H_n \rightarrow H_n$ is linear, then

$$Q_L(y, x) = (y - Lx, 0)$$

is a projection of $H_n \oplus_{\phi_n} H_n$ onto its canonical kernel and

$$\|Q_L\| \leq \max\{1, \|\phi_n - L\|\}.$$

The complementability lower bound therefore yields $\delta_n \geq c'\sqrt{n}$. Thus $\delta_n \asymp \sqrt{n}$, proving $(2) \Leftrightarrow (3)$.

Finally, H_n has dimension 3^n , so D_a has at least 3^n singular values not smaller than $|a_n|$; equivalently, $s_{3^n}(D_a) \geq |a_n|$. If $M_\omega = \sum_{j \geq 1} s_j(D_a)/j < \infty$, then

$$M_\omega \geq s_{3^n}(D_a) \sum_{j=1}^{3^n} \frac{1}{j} \geq |a_n| n \log 3.$$

It follows that $\sup_n \sqrt{n}|a_n| < \infty$. If instead $s_j(D_a) \leq C/\log(j+1)$, then

$$|a_n| \leq s_{3^n}(D_a) \leq \frac{C}{\log(3^n + 1)} \leq \frac{C'}{n},$$

and the same conclusion follows. Apply $(3) \Rightarrow (1)$. □

5 Verification and limitations

The block criterion is two-sided and uses only the standard quasilinear model of a twisted sum. The upper estimate $\delta_n \leq \sqrt{n}/2$ is derived directly from the displayed ELP recursion. The cited complementability estimate is used only for the reverse bound and hence only for the exact characterization; the positive Macaev and logarithmic-decay conclusions require only the direct upper estimate.

The restriction is substantial: D_a must be scalar on the canonical ELP blocks, and only lifting through this particular ELP extension is proved. Neither arbitrary operators in the Macaev ideal nor arbitrary twisted Hilbert extensions are covered. The canonical Ω_{ELP} is not equivalent to a centralizer [2, Proposition 3.1], so the theorem is not a restatement of the source paper’s centralizer calculation.

A bounded search of the exact question, its title, Macaev/liftability phrases, and the 2026 ELP paper found no claimed resolution of the universal endpoint problem or this aligned-block characterization. Human novelty review remains recommended.

References

- [1] F. Cabello Sánchez and R. García, *The Twisted Hilbert Space Ideals*, Integral Equations Operator Theory 94 (2022), article 21; arXiv:2112.03852.
- [2] J. M. F. Castillo and W. H. G. Corrêa, *A twisted Hilbert space not isomorphic to its dual*, arXiv:2603.22484 (2026).
- [3] P. Enflo, J. Lindenstrauss, and G. Pisier, *On the “three space problem”*, Math. Scand. 36 (1975), 199–210.